

SEAN M. TOMLIN

seanmtomlin@gmail.com · www.seanmtomlin.com · github.com/seantomlin
Biostatistician · Causal Inference · Real World Evidence · Health Policy Statistics

EDUCATION

The Ohio State University

Columbus, Ohio

Doctor of Philosophy in Biostatistics

August 2020 – August 2025

Dissertation: "Estimating Causal Effects from Observational Data in Quasi-Experimental Designs"

Advisor: Bo Lu, Ph.D.

Specializations: Causal inference, propensity scores, quasi-experimental methods, difference-in-differences, repeated measures, policy evaluation, complex survey design, constrained optimization.

The Ohio State University

Columbus, Ohio

Master of Science in Statistics

August 2020 – May 2022

Wright State University

Dayton, Ohio

Bachelor of Science in Statistics

August 2016 – May 2020

Minor in Mathematics · Summa Cum Laude with Honors

PROFESSIONAL APPOINTMENTS

2025–	Biostatistician, Medpace, Inc., Cincinnati, Ohio
2024–2025	Dissertation Year Fellow, Division of Biostatistics, The Ohio State University
2023–2024	Graduate Research Associate, College of Public Health, The Ohio State University
2020–2023	Graduate Fellow, Division of Biostatistics, The Ohio State University
2020	Research Intern, Hypersonic and Airbreathing Propulsion Branch, Research Directorate National Aeronautics and Space Administration, Langley Research Center, Hampton, Virginia
2019	Statistician Intern, Scientific Test & Analysis Techniques Center of Excellence Air Force Institute of Technology, Wright-Patterson Air Force Base, Ohio

RESEARCH INTERESTS

Causal inference from observational data; quasi-experimental designs; difference-in-differences; propensity score methods; complex survey design; health policy evaluation; clinical trials methodology; real-world evidence.

PUBLICATIONS

WORKING PAPERS

[†]Tomlin, S., Andridge, R., and Lu, B. (In preparation). Characterizing retrospective constraints of survey designs for estimating population-level causal effects. *Student Paper Competition Award, American Statistical Association (Survey Research Methods, Government Statistics, and Social Statistics Sections), Joint Statistical Meetings, Boston, MA, 2026.*

Tomlin, S., and Lu, B. (In preparation). Counterpart statistics in the matched difference-in-differences design.

PRESENTATIONS

CONTRIBUTED ORAL PRESENTATIONS

- 2026 "Characterizing Retrospective Constraints of Survey Designs for Estimating Population-level Causal Effects." Joint Statistical Meetings, Boston, Massachusetts.
- 2025 "Counterpart Statistics in the Matched Difference-in-Differences Design." Joint Statistical Meetings, Nashville, Tennessee.
- 2025 "Revisiting Covariate Adjustment in Parametric Difference-in-Differences." ENAR Spring Meeting, New Orleans, Louisiana.
- 2024 "On the Matched Difference-in-Differences Design with Counterparts." Joint Statistical Meetings, Portland, Oregon.
- 2023 "Matching Designs for Causal Inference in Health Policy Evaluation." Research Day, College of Public Health, The Ohio State University.

POSTERS AND SPEED PRESENTATIONS

- 2025 "Counterpart Statistics in the Matched Difference-in-Differences Design." American Causal Inference Conference, Detroit, Michigan.
- 2024 "On the Matched Difference-in-Differences Design with Counterparts." Golden Anniversary Celebration, Department of Statistics, The Ohio State University.
- 2024 "Revisiting the Design and Analysis of Parametric Difference-in-Differences." Departmental Poster Session, Department of Statistics and Division of Biostatistics, The Ohio State University.
- 2023 "Negative Control Adaptive Randomization Inference for Matched Observational Studies." Departmental Poster Session, Department of Statistics and Division of Biostatistics, The Ohio State University.

RESEARCH EXPERIENCE

CLINICAL RESEARCH

Medpace, Inc. — Clinical Contract Research Organization

*Cincinnati, Ohio
2025 – Present*

Lead statistical and programming activities across Phase 1-3 industry-sponsored clinical trials, including developing statistical analysis plans, selecting appropriate methodologies, programming analyses, and preparing statistical methods sections for clinical reports. Support clinical development planning and collaborate with data management and medical writing teams.

GRADUATE RESEARCH

The Ohio State University — Division of Biostatistics

*Columbus, Ohio
2021 – 2025*

Advisor: Bo Lu, Ph.D.

Dissertation research on causal inference methods for quasi-experiments, with focus on difference-in-differences designs, nonnegative outcomes in healthcare utilization, and retrospective constraints from survey design.

Graduate Research Associate in collaboration with the Center for Injury Research and Policy at Nationwide Children's Hospital; contributed to applied studies in public health and injury epidemiology.

GRANT APPLICATIONS

2023 NIH F31 Predoctoral Fellowship. "Causal inference for policy evaluations with individual-level survey data: Estimating the impact of Tobacco 21 policies on adolescent smoking behavior." Not funded.

STATISTICAL CONSULTING

2025 Denison University. Statistical analysis and reproducible report for a repeated measures experimental design. Granville, Ohio. July 2025 – December 2025.

HONORS AND AWARDS

2026 †Student Paper Award, SRMS/GSS/SSS Consortium of Sections, Joint Statistical Meetings
2025 Craig Cooley Memorial Prize, Department of Statistics, The Ohio State University
2024 Student Travel Award, Survey Research Methods Section, Joint Statistical Meetings
2024–2025 Dissertation Year Fellowship, The Ohio State University
2022–2023 College of Public Health Fellowship, College of Public Health, The Ohio State University
2020–2022 Susan Huntington Dean's Distinguished University Fellowship, The Ohio State University
2020 University Honors Scholar, Wright State University
2016–2020 Competitive Honors Scholarship, Wright State University
2016–2020 Valedictorian Scholarship, Wright State University

MENTORING

July 2025 Yuan Wang, Oberlin College. Ohio Five Summer Undergraduate Research Experience.
July 2024 Syad Rafey, Kenyon College. Ohio Five Summer Undergraduate Research Experience.
July 2023 Lynn Nguyen, Denison University. Ohio Five Summer Undergraduate Research Experience.

PROFESSIONAL SERVICE

LEADERSHIP AND OUTREACH

2023–2025 Journal Club Leader, Ohio Five OSU Summer Undergraduate Research Experience
2023–2024 President, Statistics and Biostatistics Graduate Student Association, The Ohio State University
2023–2024 Peer Mentor, Statistics and Biostatistics Graduate Student Association, The Ohio State University
2023–2024 Volunteer Judge, Ohio State Science Fair
2023–2024 Volunteer, ASA DataFest, The Ohio State University

PEER REVIEW

Annals of Applied Statistics
Observational Studies

PROFESSIONAL MEMBERSHIPS

2022– American Statistical Association
2024– Eastern North American Region (ENAR) of the International Biometric Society
2025– Society for Causal Inference